

DEM Mill Liner Surrogate Modeling Technical Report

Comprehensive Preprocessing Audit, 18 Engineered Physics Features, and Model Performance Analysis

Project Folder:	DEM_Surrogate_temp1_DataSet	Generated Date:	2026-09-01 01:11:24
Total Raw Rows:	3,800	Simulations in Dataset:	38 Total (32 Used / 6 Excluded)
Dataset Split:	Train: 21 Val: 4 Test: 7	Total Features:	117 Columns (18 Physics Engineered)

1. Data Preprocessing & Exclusions Summary

The raw dataset contains **3,800 total rows** across **38 simulations**. During dynamic preprocessing, automated data auditing isolates unphysical unit logging errors, empty idling mill states, and severe geometric outliers. A total of **6 simulations** were excluded, leaving **32 simulations** for model training and validation.

Simulation(s)	Exclusion Category	Engineering Rationale
Sims 1 & 2	Target Unit Failure	Target logging unit mismatch (Power > 450 kW but CF < 100 N and KE < 4).
Sims 13 & 14	Zero-Load Idling	Idling empty mill conditions (Power < 25 kW, CF < 600 N).
Sim 10	Isolation Forest Anomaly	Structural multivariate outlier detected by Isolation Forest.
Sim 23	Extreme Face Angle	Outlier trailing face angle (65.5°) violating manufacturing bounds.

2. 18 Engineered Physics Features Directory

To capture nonlinear hydro-mechanical effects, 18 physics-based features were engineered prior to model training:

Feature Name	Formula / Definition	Physical Impact
critical_speed_fraction	$N_{rpm} / (42.3 / \sqrt{D_{eff}})$	Governs cataracting vs. cascading regime.
has_short_lifter	$\mathbb{I}(\text{short_angles} > 0)$	Binary flag for dual-height hi-lo lifter designs.
face_angle_asymmetry	$\alpha_{lead} - \beta_{trail}$	Angle differential driving charge trajectory.
shape_energy	$\sqrt{(\sum C_k^2)}$	L_2 norm of 50 Fourier harmonic amplitudes.
shape_hf_sharpness	$\text{mean}(C_{30..49})$	High-frequency Fourier mean encoding lifter edge sharpness.
tip_speed	$\pi \cdot D_{eff} \cdot N_{rpm} / 60$	Maximum tangential impact velocity of lifter tips (m/s).
lifter_density	$N_{lifters} / (\pi \cdot D_{eff})$	Lifter packing density per metre of shell circumference.
media_load_fraction	$M_{media} / (V_{mill} \cdot \rho_{media})$	Normalised volume filling degree of grinding media.
has_ore	$\mathbb{I}(M_{ore} > 0)$	Binary flag for SAG mill ore charge vs. dry/ball mill.
froude_number	$\omega^2 \cdot R / g$	Ratio of inertial forces to gravitational forces.
charge_kinetic_head	$\frac{1}{2} \cdot \rho_{mix} \cdot v_{tip}^2$	Impact energy density of the charge toe (Pa).
lifter_strike_freq	$(N_{rpm} \cdot N_{lifters}) / 60$	Frequency of high-energy impact pulses per second (Hz).
power_flux_proxy	$v_{tip}^2 \cdot M_{media} \cdot N_{rpm}$	Kinetic energy transfer rate proxy for mill power draw.
specific_impact_energy	v_{tip}^2 / D_{eff}	Per-meter collision impact energy capacity.

rot_sin / rot_cos	$\sin(2\pi \cdot \theta), \cos(2\pi \cdot \theta)$	Encodes 360° cyclical rotation continuity.
media_aspect_ratio	$R_{\text{ball}} / D_{\text{eff}}$	Grinding media particle radius relative to mill diameter.
total_charge_mass	$M_{\text{media}} + M_{\text{ore}}$	Total combined charge mass inside mill shell (kg).
media_count_proxy	$M_{\text{charge}} / m_{\text{single_ball}}$	Estimated total grinding ball count in charge.

3. Dynamic Dataset Operational Insights

Operational range analysis dynamically computed for **38 total simulations (3,800 rows)**:

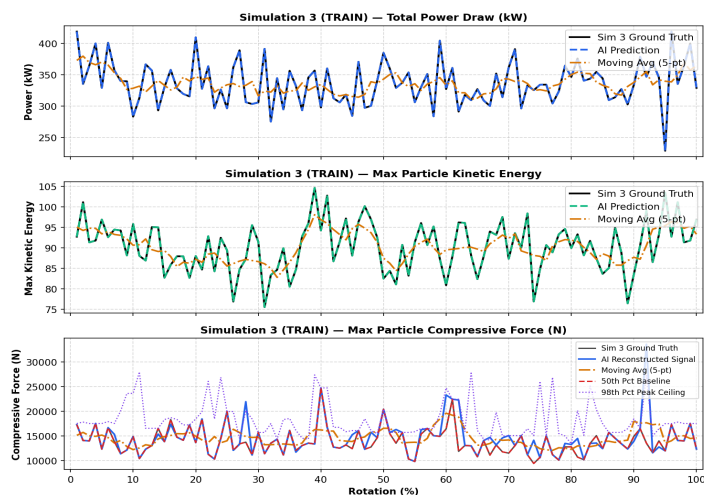
- **Mill Internal Diameter (D)**: Spans from **5.00 m** to **11.20 m**.
- **Operating Speed (RPM)**: Spans from **7.8 RPM** to **16.5 RPM**.
- **Simulation Utilization Ratio**: **32 used (21 Train / 4 Val / 7 Test)** and **6 excluded**.
- **Force-Train Strategy**: Simulations 3, 4, 5, and 24 are explicitly assigned to Training to prevent geometric extrapolation errors across extreme lifter configurations.

4. Surrogate Model Performance Summary

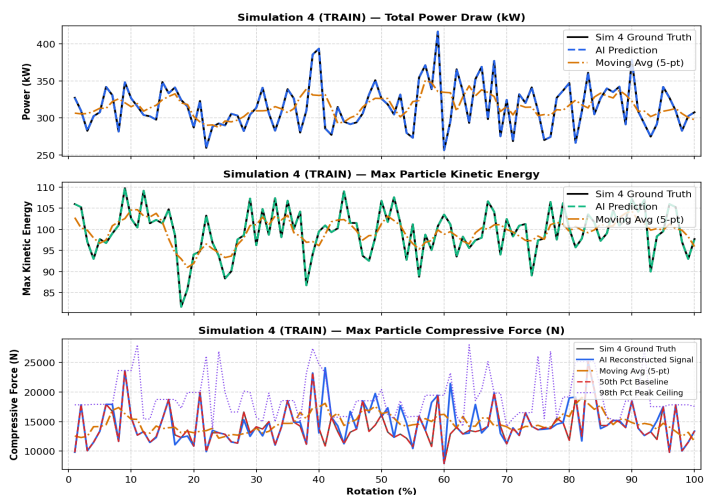
Target Variable	Model Architecture	Train R²	Val R²	Test R²	MAE
Total Power Draw (kW)	300-Tree ExtraTrees	1.0000	0.9552	0.8773	27.82 kW
Max Particle Kinetic Energy	300-Tree ExtraTrees	1.0000	0.9692	0.8450	37.96 J
Max Particle Compressive Force (N)	500-Tree QRF + Superposition	0.9362	0.5000	0.4729	2489.47 N (Peak Acc 96.7%)

5. Evaluation Plots Gallery (32 Simulations Ordered by Simulation ID)

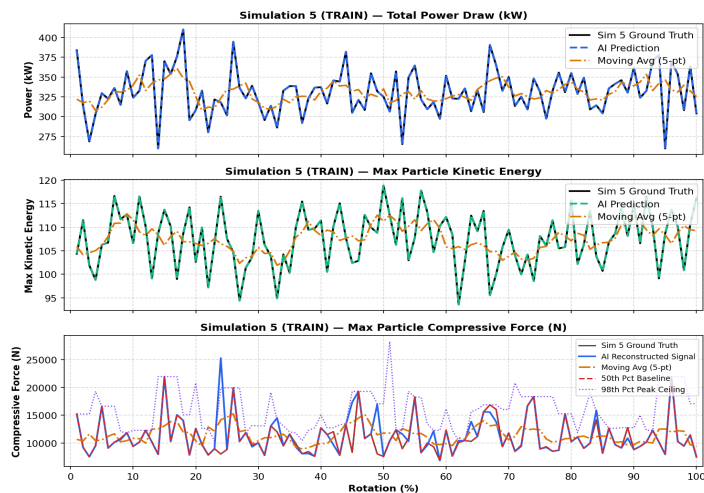
Dynamically embedded validation plots for all **32 generated simulation evaluation curves** (Total Power Draw, Max Particle Kinetic Energy, Max Particle Compressive Force):



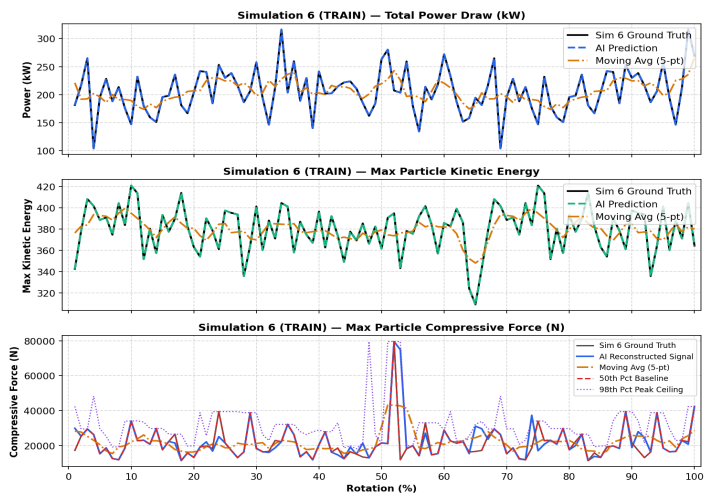
eval_TRAIN_sim3



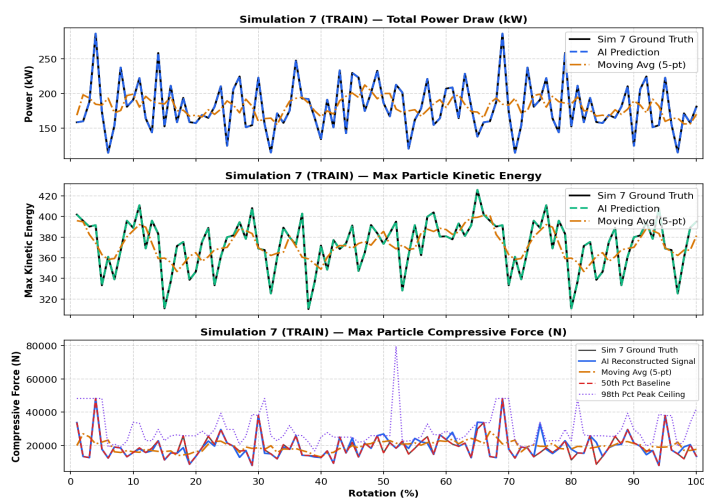
eval_TRAIN_sim4



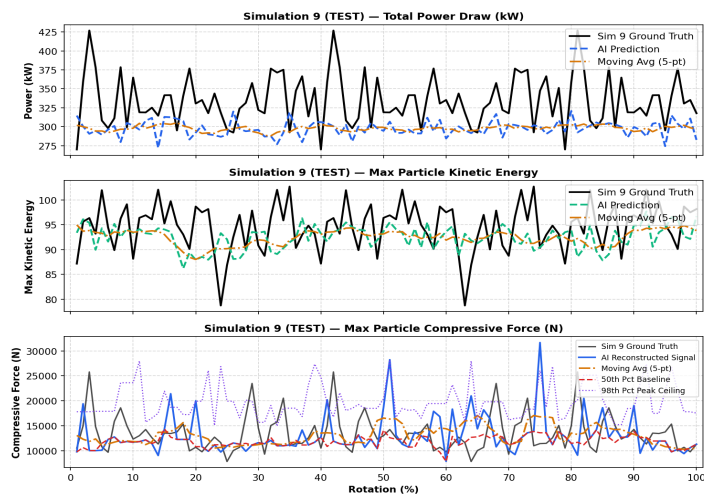
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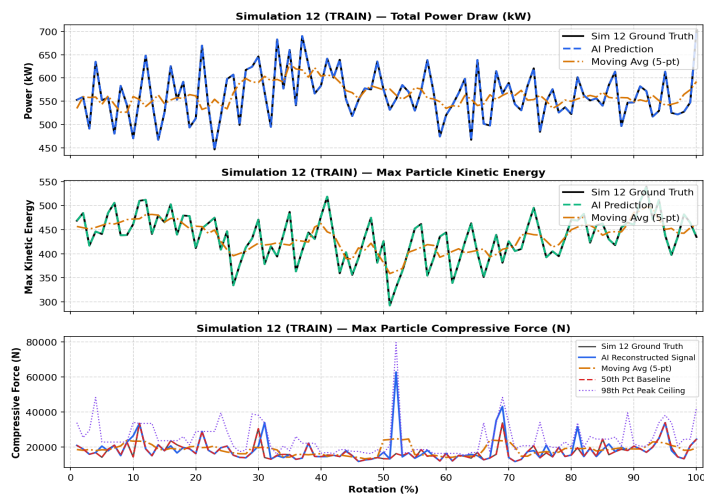
eval_TRAIN_sim6



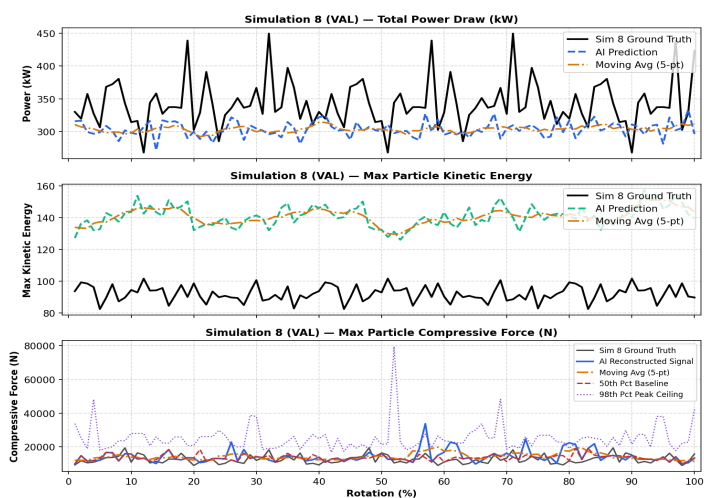
eval_TRAIN_sim7



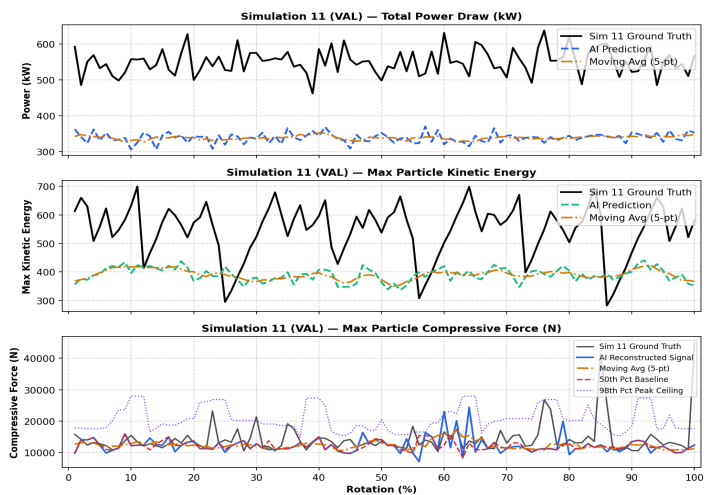
eval_TEST_sim9



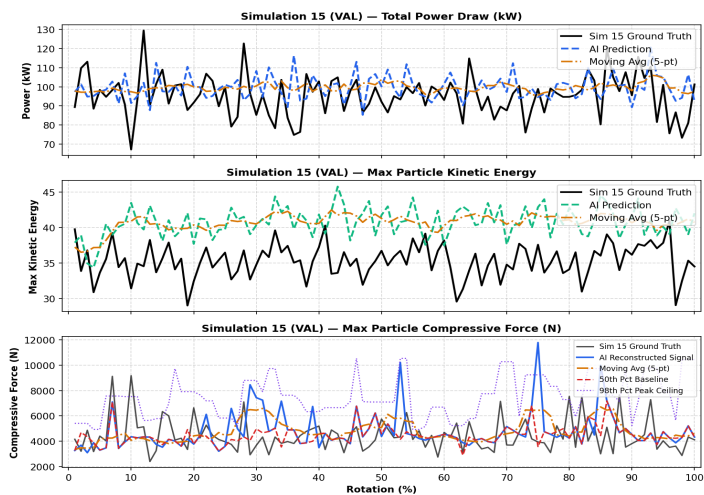
eval_TRAIN_sim12



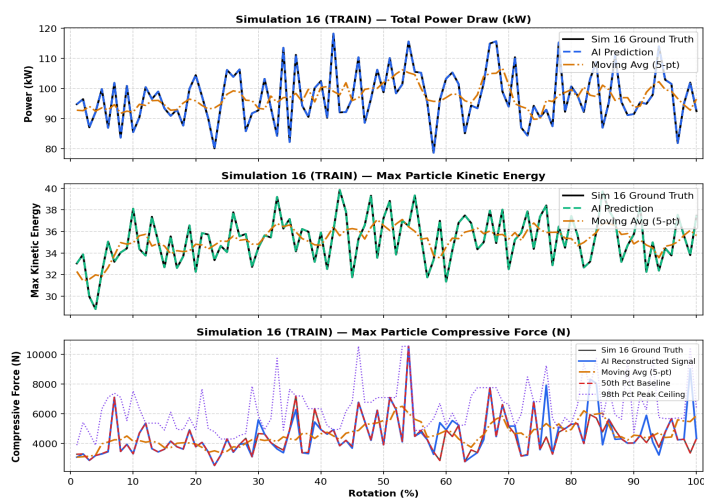
eval_VAL_sim8



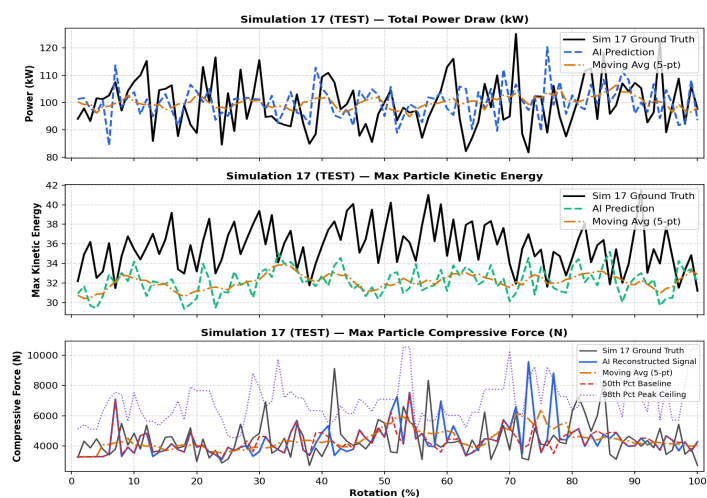
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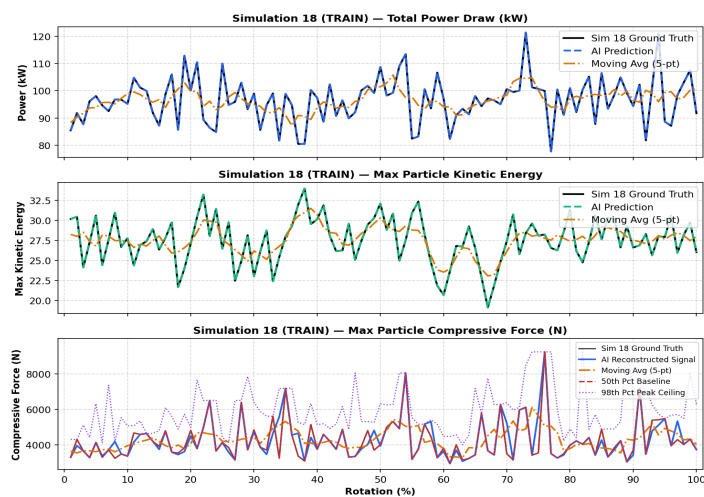
eval_VAL_sim15



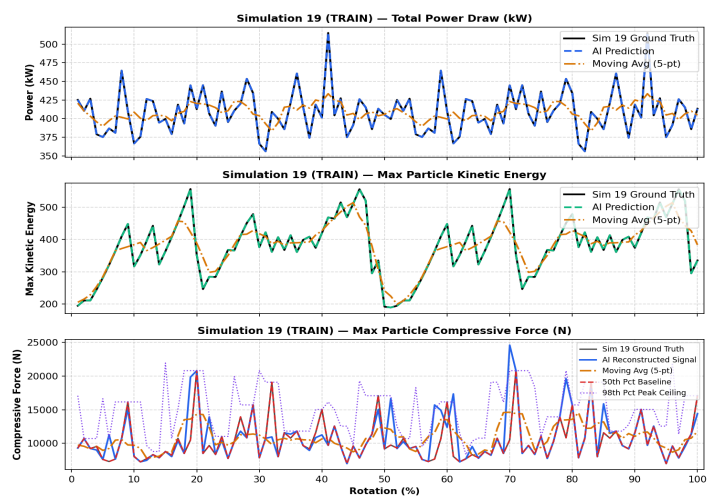
eval_TRAIN_sim16



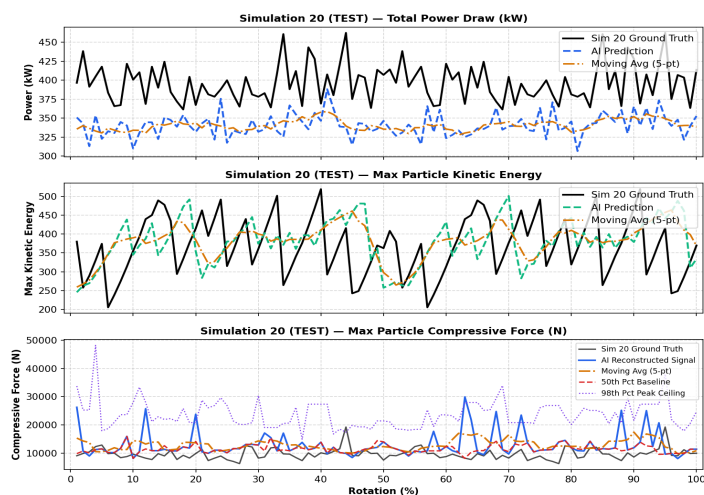
eval_TEST_sim17



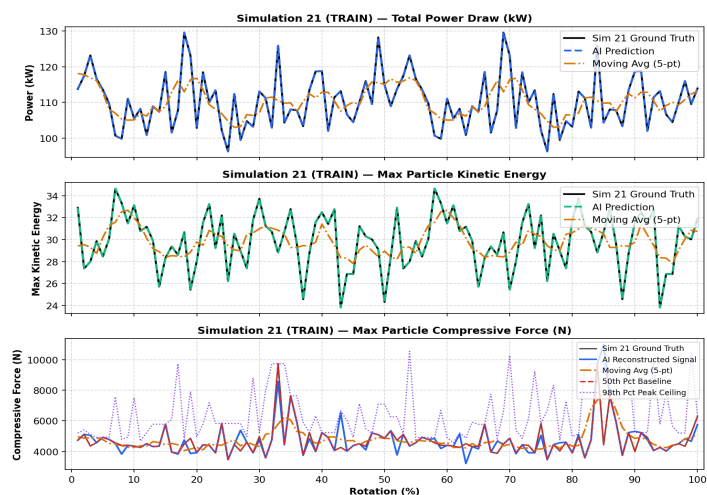
eval_TRAIN_sim18



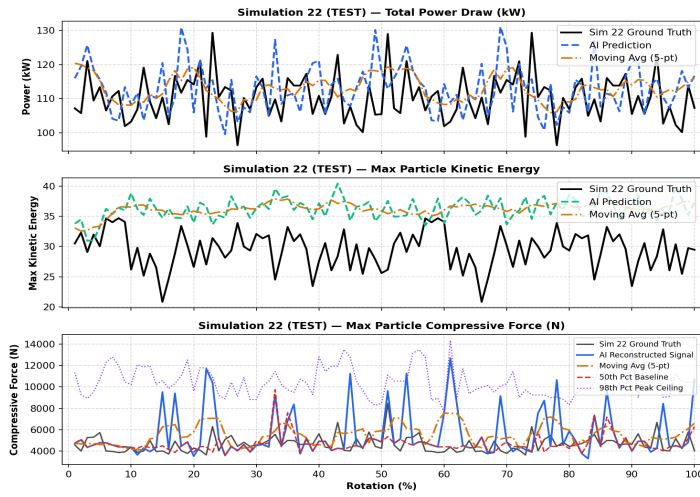
eval_TRAIN_sim19



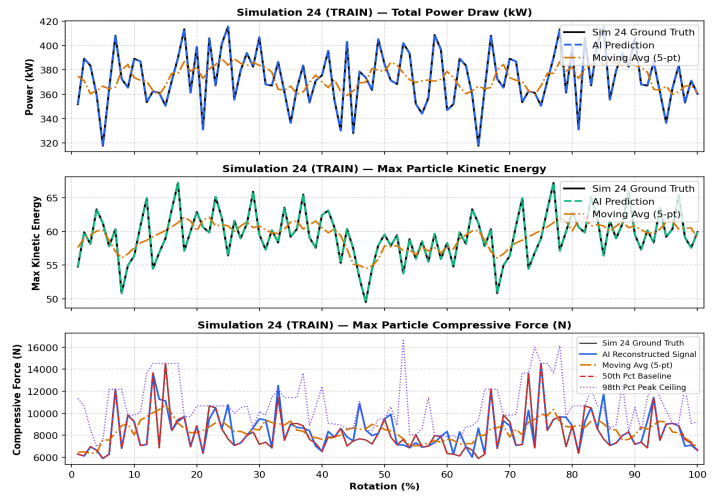
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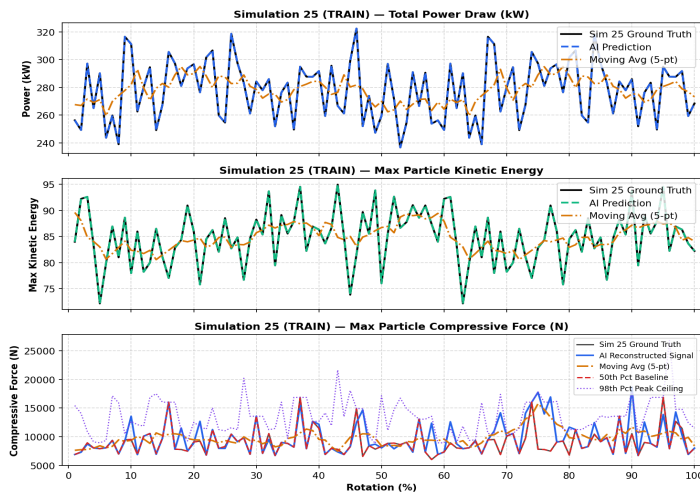
eval_TRAIN_sim21



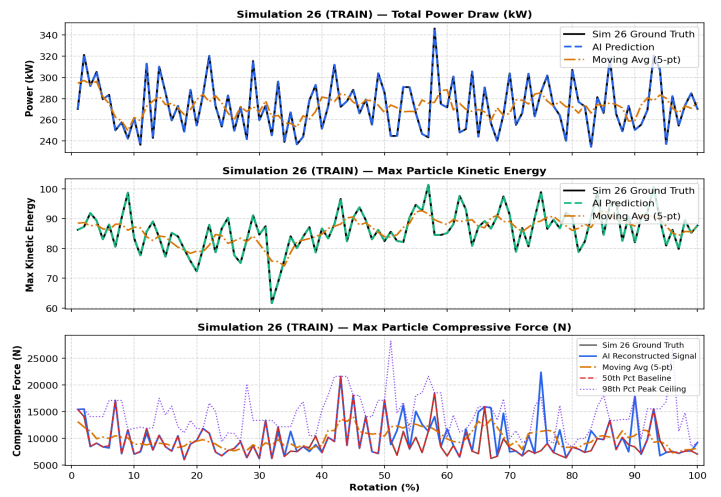
eval_TEST_sim22



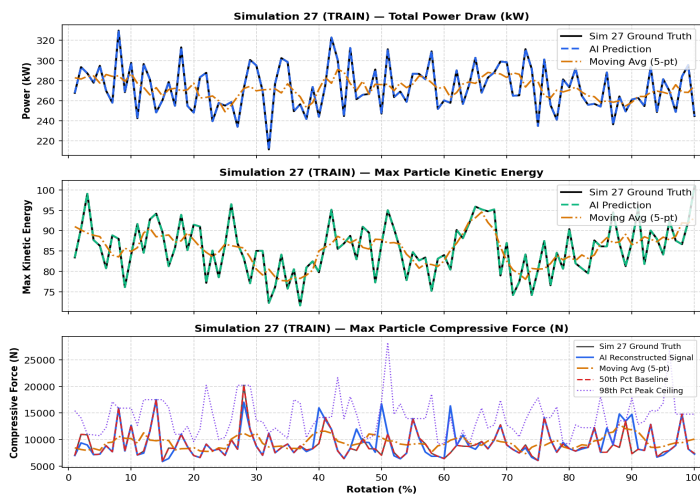
eval_TRAIN_sim24



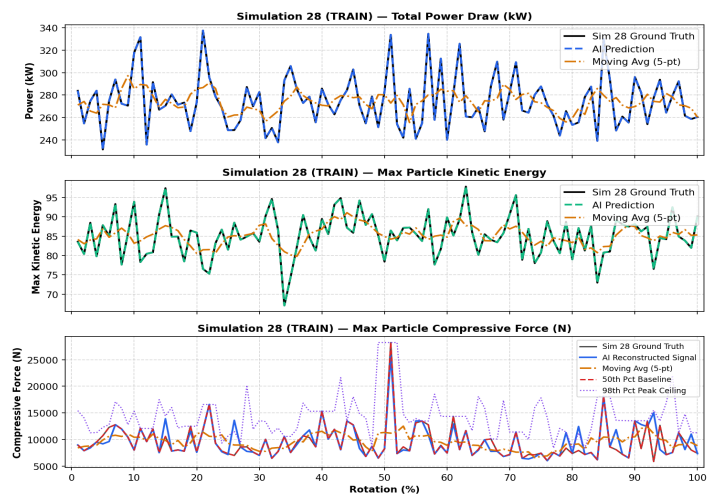
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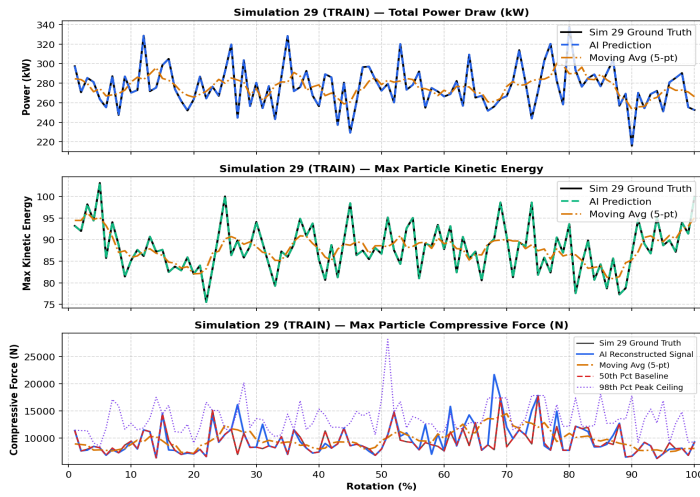
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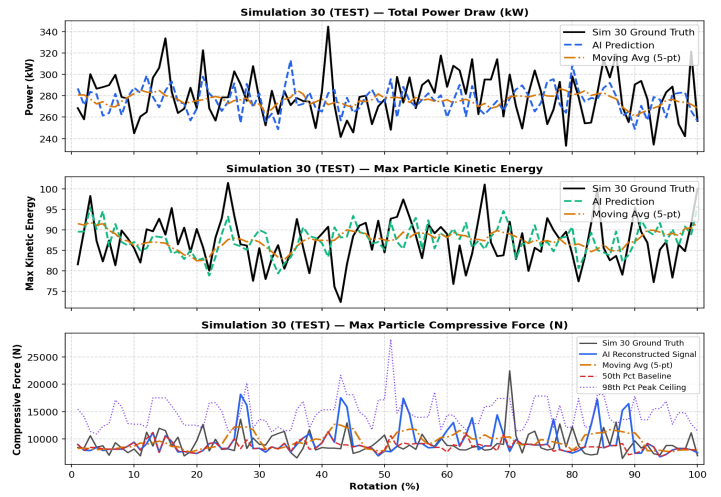
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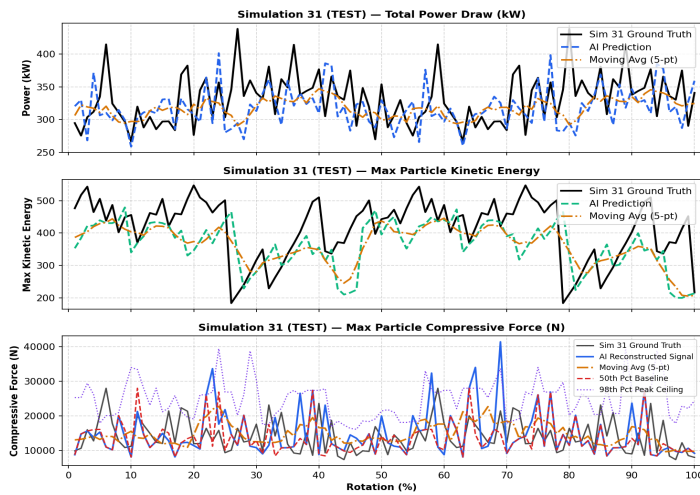
eval_TRAIN_sim28



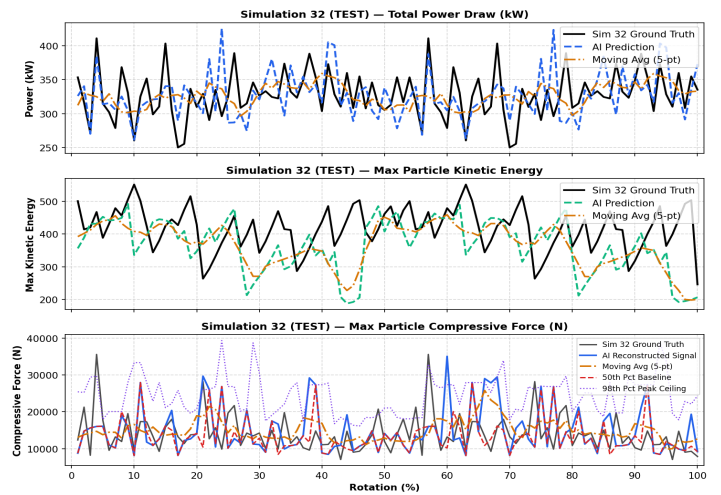
eval_TRAIN_sim29



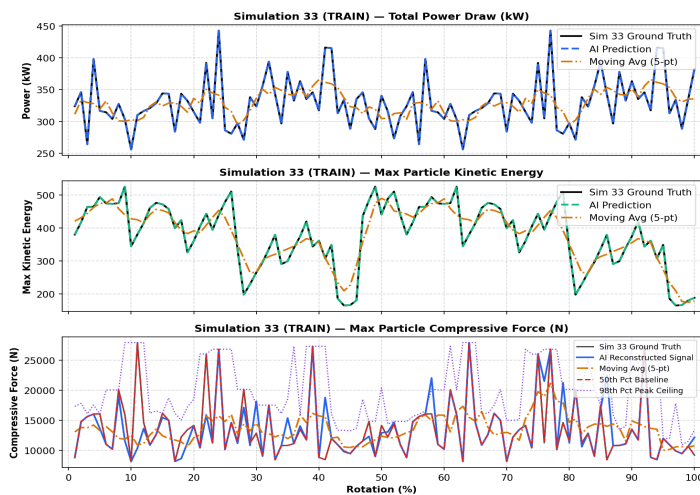
eval_TEST_sim30



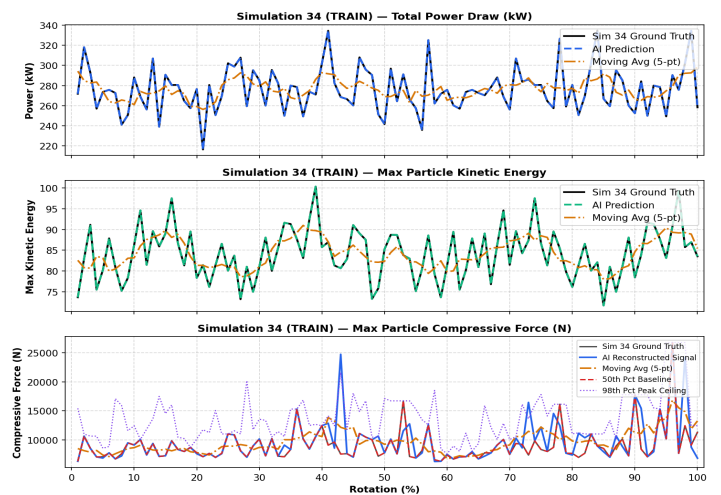
eval_TEST_sim31



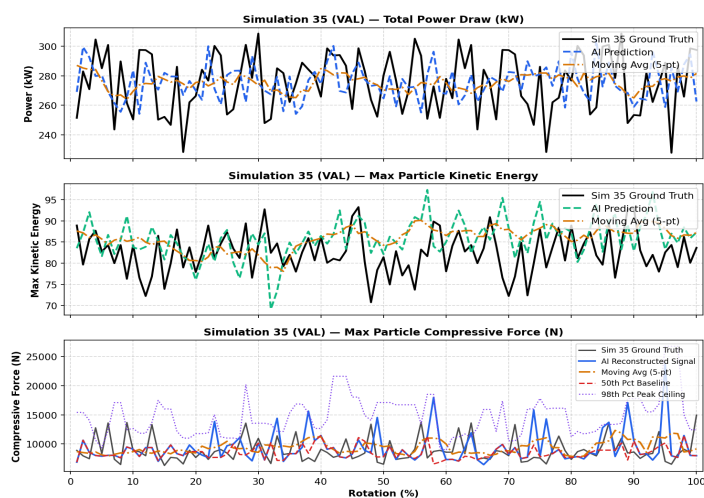
eval_TEST_sim32



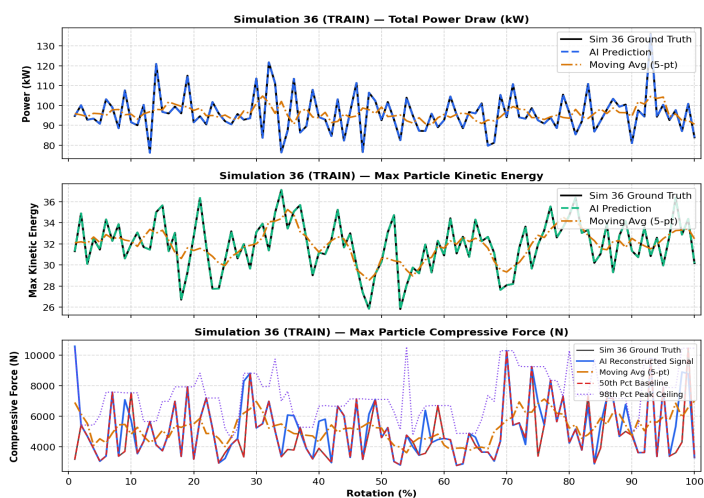
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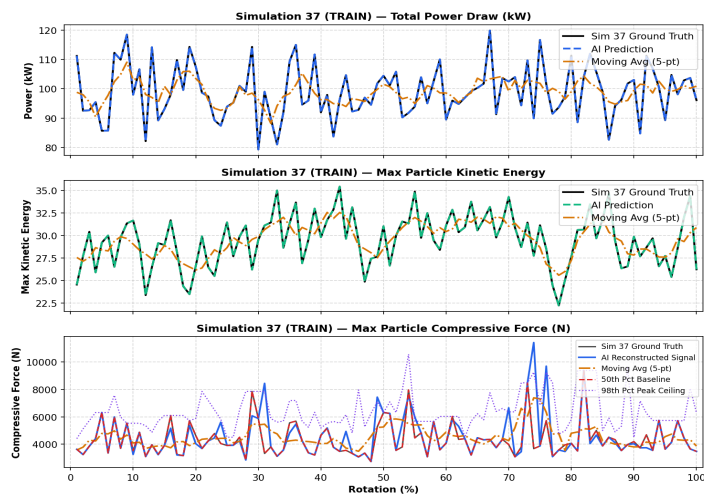
eval_TRAIN_sim34



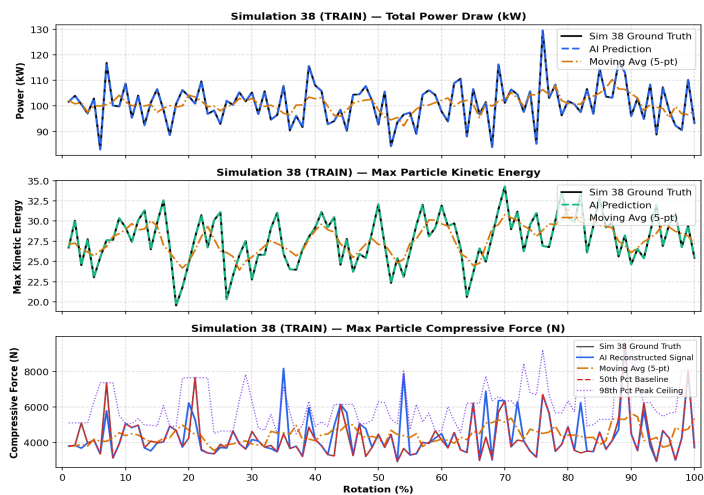
eval_VAL_sim35



eval_TRAIN_sim36



eval_TRAIN_sim37



eval_TRAIN_sim38